

IELTS Marking Calibration – Executive Overview & Requirements

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Purpose: what it takes to switch the AI IELTS marker from "built but dark" to "live for learners" — people, workload, cost, time.

1. The one-paragraph summary

The AI marking engine is built and safe, but it is **deliberately locked** until we can prove it agrees with real human examiners. To unlock it we need a **calibration set**: a bank of real IELTS Writing Task 2 essays, each independently marked by **two qualified IELTS examiners**, plus a small set of **annotated exemplars**. The engine marks the same essays, we measure agreement, and the system only goes live when the AI lands **within half a band of the human examiners on at least 80% of essays**. This is a **one-off data-build of roughly 3–4 weeks**, needing **3–4 qualified IELTS markers** for **~10–15 hours each**, at an estimated **£1,000–£1,800 all-in**. After that, a much smaller "top-up" keeps it honest over time.

2. Why this is needed (and why it's non-negotiable)

The platform marks children's and fee-paying candidates' work. A confidently-wrong band score damages a learner's exam prep and our reputableness. So the engine has a **hard gate in code** (`assertCalibrationGreen`): the live IELTS route returns "not yet calibrated" (HTTP 503) and shows **nothing** to a learner until a green calibration baseline exists. This is not optional polish — it is the trust foundation, and it's already enforced in the request path. The calibration data is the **only** thing standing between "built" and "live."

3. What we are actually building

Two assets, both produced by human markers:

ASSET	WHAT IT IS	WHY THE ENGINE NEEDS IT
Calibration set	≥50 IELTS Writing Task 2 essays, each marked independently by 2 examiners against the 4 criteria (Task Response, Coherence & Cohesion, Lexical Resource, Grammatical Range & Accuracy) on the 0–9 band scale	The engine marks the same essays; we compare AI-vs-human to prove accuracy and catch drift. This is the gate.
Exemplar bank	A small set of essays at each band (4–9), with a short expert annotation explaining <i>*why*</i> it sits at that band per criterion	These are the "anchors" injected into the AI at marking time — the single biggest lever on marking accuracy. Currently zero exist.

Important: the essays themselves can be **synthetic** (the platform generates them to a target band — copyright-clean, no real-student data). The **marks must be human** — we never calibrate the AI against another AI's opinion. Markers grade synthetic essays exactly as they would real ones.

4. The numbers — what we recommend you commission

We recommend marking slightly **above** the gate minimums so the baseline is robust, not borderline.

ITEM	GATE MINIMUM	RECOMMENDED COMMISSION	WHY ABOVE MINIMUM
Essays in the calibration set	50	60	Buffer against dropped/disagreed essays
Bands covered	4–9	4–9 (all six)	The gate needs ≥ 2 populated bands; we want the full range
Essays per band	5	10 per band	Double the floor → statistically trustworthy per-criterion figures
Marks per essay	2 (double-marking)	2	Inter-marker agreement is how we trust the "human truth"
Exemplars	≥ 1 per band 5–8	2 per band, 4–9 (12 total)	A spread of anchors, expert-annotated
Within-half-band agreement floor	0.80	target ≥ 0.85	Aim above the floor so the build passes comfortably

Resulting workload

- **Calibration marks:** 60 essays $\times$ 2 markers = **120 independent marks**
- **Adjudication:** where the 2 markers disagree by >1 band (expect $\sim 15\text{--}25\%$ → $\sim 12\text{--}15$ essays), a **third (senior) mark** resolves it = **~ 15 adjudication marks**
- **Exemplar annotations:** 12 exemplars, each written up by the senior examiner = **12 annotations**
- **Total marker-events:** **~ 135 marks + 12 annotations**

5. How many markers, and who they need to be

Recommended team: **3 qualified IELTS markers + 1 senior lead** (the lead can be one of the 3, wearing two hats, if budget is tight → effectively a 3-person team).

ROLE	HOW MANY	WHAT THEY DO	SKILL SET REQUIRED
IELTS Writing markers	3	The bulk double-marking — each marks ~40 essays (so every essay gets 2 of the 3)	Current or recent certified IELTS examiner (or equivalent: DELTA/CELTA + IELTS-prep teaching experience marking to the public band descriptors). Native or near-native English. Reliable, detail-oriented, comfortable with a structured online form.
Senior lead examiner	1	Adjudicates disagreements, writes the 12 exemplar annotations, sanity-checks the set. (Can be your strongest marker.)	As above plus examiner-trainer / senior-examiner experience and the judgement to write authoritative per-band annotations.

Why 3, not 2: double-marking needs at least 2, but with only 2 markers one person's bias dominates every essay and there's no one independent to adjudicate. Three gives genuine inter-marker agreement, spreads the load, and provides the adjudicator. Four would be slightly more robust but isn't necessary for a 60-essay set.

Why qualified examiners specifically: the whole point is to prove the AI matches *examiner* judgement. Marks from a non-examiner are not a valid reference — they'd give us a green light we couldn't trust. This is the one place we cannot cut the skill bar.

6. Per-person workload & time

	PER MARKER (×3)	SENIOR LEAD
Essays to mark	~40 calibration essays	~40 (as a marker) + ~15 adjudications + 12 exemplar annotations
Time per essay	~12–18 min (band + 4 criterion sub-marks + brief evidence notes via the form)	same, + ~20 min per exemplar annotation
Total time	~10–12 hours each	~15–18 hours
Spread	Comfortably part-time over 1–2 weeks	over 2 weeks

For a marker, this is **less than two normal working days**, done flexibly from home.

7. Cost to you

Marking rates for freelance IELTS examiners vary by market and seniority. The figures below are **planning estimates** — confirm the actual rate when you recruit (the platform pays a configurable **pence-per-approved-script** rate, so you set it).

SCENARIO	RATE / SCRIPT	CALIBRATION (120 MARKS)	ADJUDICATION (15)	EXEMPLARS (12)	LEAD OVERSIGHT	TOTAL
Lean	£4	£480	£60	£120	(lead is a marker)	~£660
Base (recommended)	£6	£720	£90	£180 (£15 ea)	£150	~£1,140
Premium (senior examiners, higher rate)	£9	£1,080	£135	£300 (£25 ea)	£300	~£1,815

Plan for ~£1,000–£1,500 all-in for the one-off calibration build. This is a **one-time** spend — once green, the engine is live. (For comparison: it's roughly one founding-school pilot fee, and it unlocks the entire IELTS product.)

Ongoing cost (after go-live)

A small **monthly top-up** keeps calibration honest as the model/prompt evolves: ~10–15 fresh double-marked essays/month ≈ **£150–£250/month**. Optional but recommended — it's the early-warning system if marking quality drifts.

8. End-to-end process (8 steps)

1. GENERATE essays	→ platform's synthetic-essay generator produces 60 IELTS WT2 essays across bands 4–9 (copyright-clean, no real-student data)
2. RECRUIT + ACTIVATE	→ 3 markers + 1 lead onboarded into /admin/marker-drive; IELTS added to their profile; consent + pay rate set
3. CREATE BATCH	→ a "calibration" batch (board=IELTS, paper=WT2) loaded with the 60 essays

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| 4. ASSIGN (double) | → each essay assigned to 2 of the 3 markers |
| 5. MARK | → markers grade via their queue: per-criterion bands + brief evidence notes (the structured form) |
| 6. QA + ADJUDICATE | → system computes inter-marker agreement; >1-band disagreements go to the senior lead for a deciding mark; the agreed mark becomes the "human truth" |
| 7. PROMOTE | → QA-passed marks promoted into the calibration set; the senior lead's annotated exemplars promoted into the knowledge pack |
| 8. CALIBRATE + GO-LIVE | → engine marks the set, agreement is measured; when within-half-band ≥ 0.80 (we target 0.85) the gate flips green and the IELTS route opens to learners |

You (or an ops person) drive steps 1–4 and 7–8 from the admin console; the markers do 5–6.

9. Timeline

WEEK	ACTIVITY	OWNER
Week 1	Generate essays · recruit + onboard markers · create batch + assign	You / ops
Weeks 1–2	Markers complete double-marking (~10 hrs each, flexible)	Markers
Week 3	QA, adjudication, exemplar annotations	Senior lead
Week 3–4	Promote → run calibration → green baseline → go live	You / ops

Realistic end-to-end: 3–4 weeks, gated mostly by marker availability, not by us. A motivated team could compress to ~2 weeks.

10. What you need to decide / provide

1. **Approve the budget band** (~£1,000–£1,500 base case).
2. **Set the pence-per-script rate** in the admin console (drives all payments automatically).
3. **Source 3–4 markers** — your network, an IELTS-examiner freelance pool, or a marking agency. (The send-out brief, [IELTS-marker-brief.md](#), is written to recruit them.)
4. **Confirm Academic-only** for the MVP (per decision OQ-2) — General Training is a fast-follow that needs its own small set.

Everything else — the generator, the marker console, the QA maths, the gate — is built and waiting.